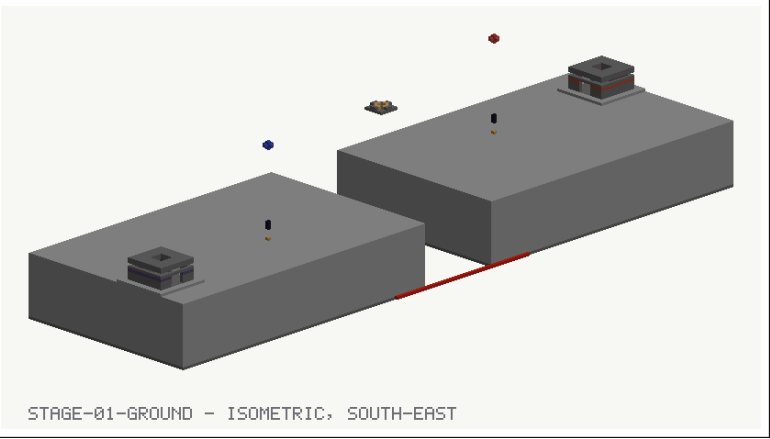


How a PGM Map Gets Made

A map is made by two parties, and the line between them moves at every stage

One board, seven stages

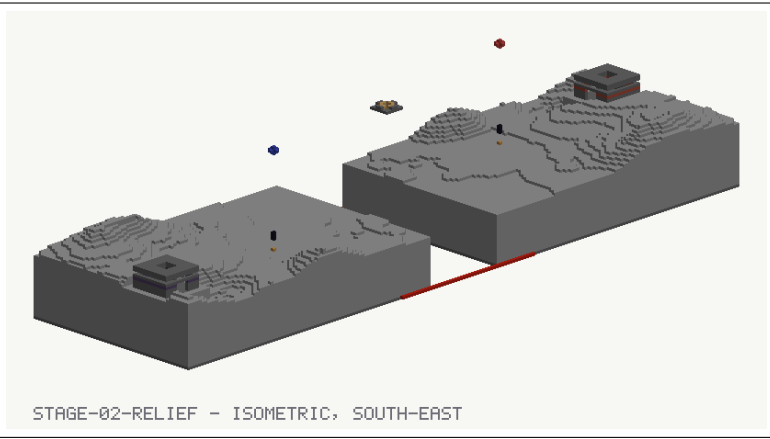
Each picture is the same map exported again with one more instrument switched on. Nothing between them is drawn by hand.



STAGE-01-GROUND - ISOMETRIC, SOUTH-EAST

1 **Ground**

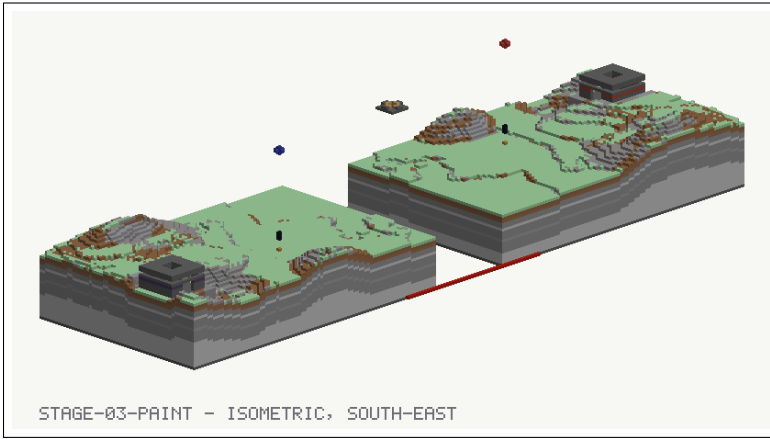
The plan compiled. Flat, grey, rectangular.



STAGE-02-RELIEF - ISOMETRIC, SOUTH-EAST

2 **Relief**

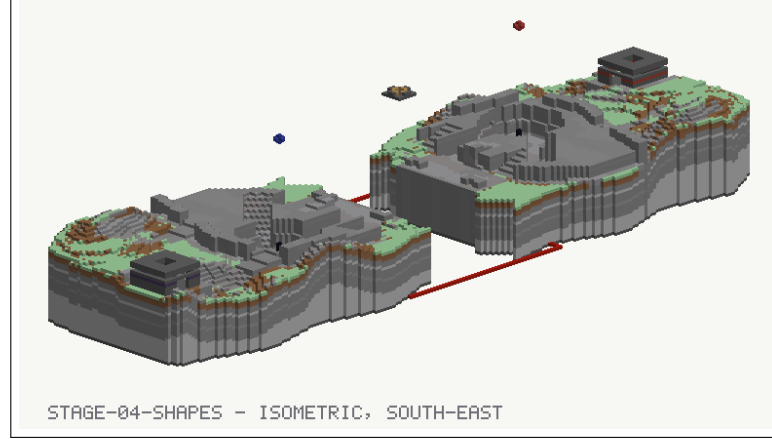
Marks pinned; the solve fills between them.



STAGE-03-PAINT - ISOMETRIC, SOUTH-EAST

3 **Paint**

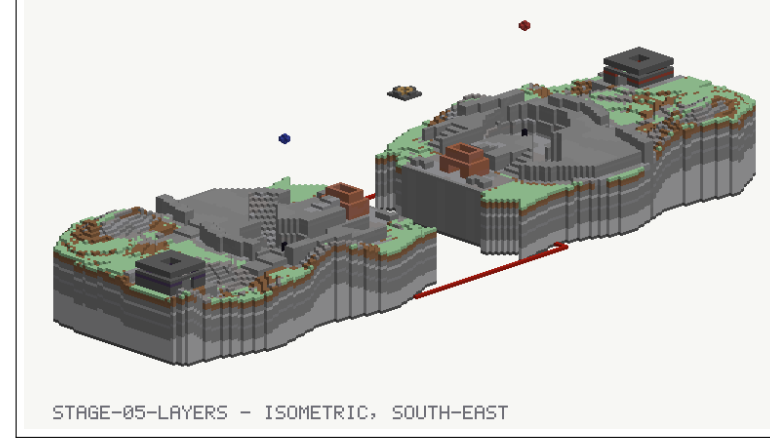
A theme per bucket, banded by the ground's angle.



STAGE-04-SHAPES - ISOMETRIC, SOUTH-EAST

4 **Shapes**

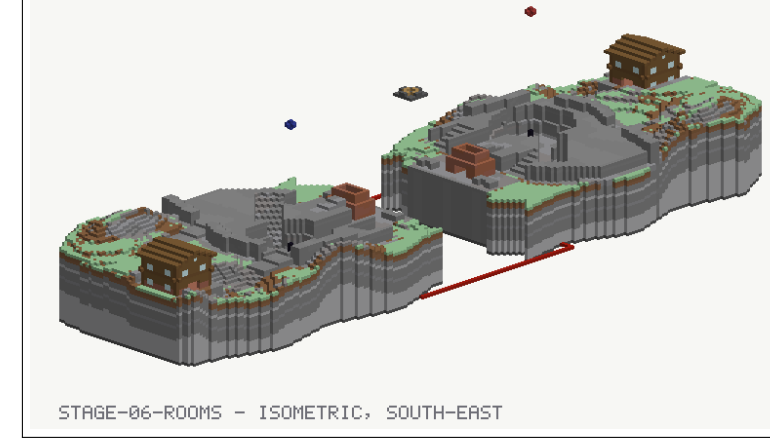
Outlines deformed and bent; landforms added.



STAGE-05-LAYERS - ISOMETRIC, SOUTH-EAST

5 **Layers**

Storeys. Depth in the literal sense.



STAGE-06-ROOMS - ISOMETRIC, SOUTH-EAST

6 **Rooms**

The shells a spawn and a wool are stamped as.



STAGE-07-DRESSED - ISOMETRIC, SOUTH-EAST

7 **Dressed**

Trees, boulders and paths, each one placed.

A map is four documents

Not four drafts. Four descriptions of the whole map at four grains, each owned by a different tool.

Level	What it states
The board	rectangles on a coarse grid, what each is for, where the objectives sit
The ground	real geometry at block resolution: outlines, heights, relief, paint
The play	teams, spawns, build regions, objectives and how they are captured
The map	the voxel world and the <code>map.xml</code> a server loads

The flow is one way. Nothing reads back up, and a finished map cannot be decompiled into the plan that would have produced it.

Who decided this?

The question the whole paper asks, stage by stage. **Blue** is the studio. **Orange** is the agent driving it.

Stage	Studio	Agent
Arrangement	hub, spawn, wool approaches, frontline, land budget	which board, and everything the composer does not state
Elevation	the solved field	the marks, the grain, the pushes
Paint	which block on which cell	the theme, the axis, where the bands cut
Structures	storey heights, roof pitch, door side	which style, which seat from the offered set
Dressing	the legal seats	which seats to take
Objectives	regions, filters, monuments	what the map is played for
Refusal	what may not be built	what to do about it

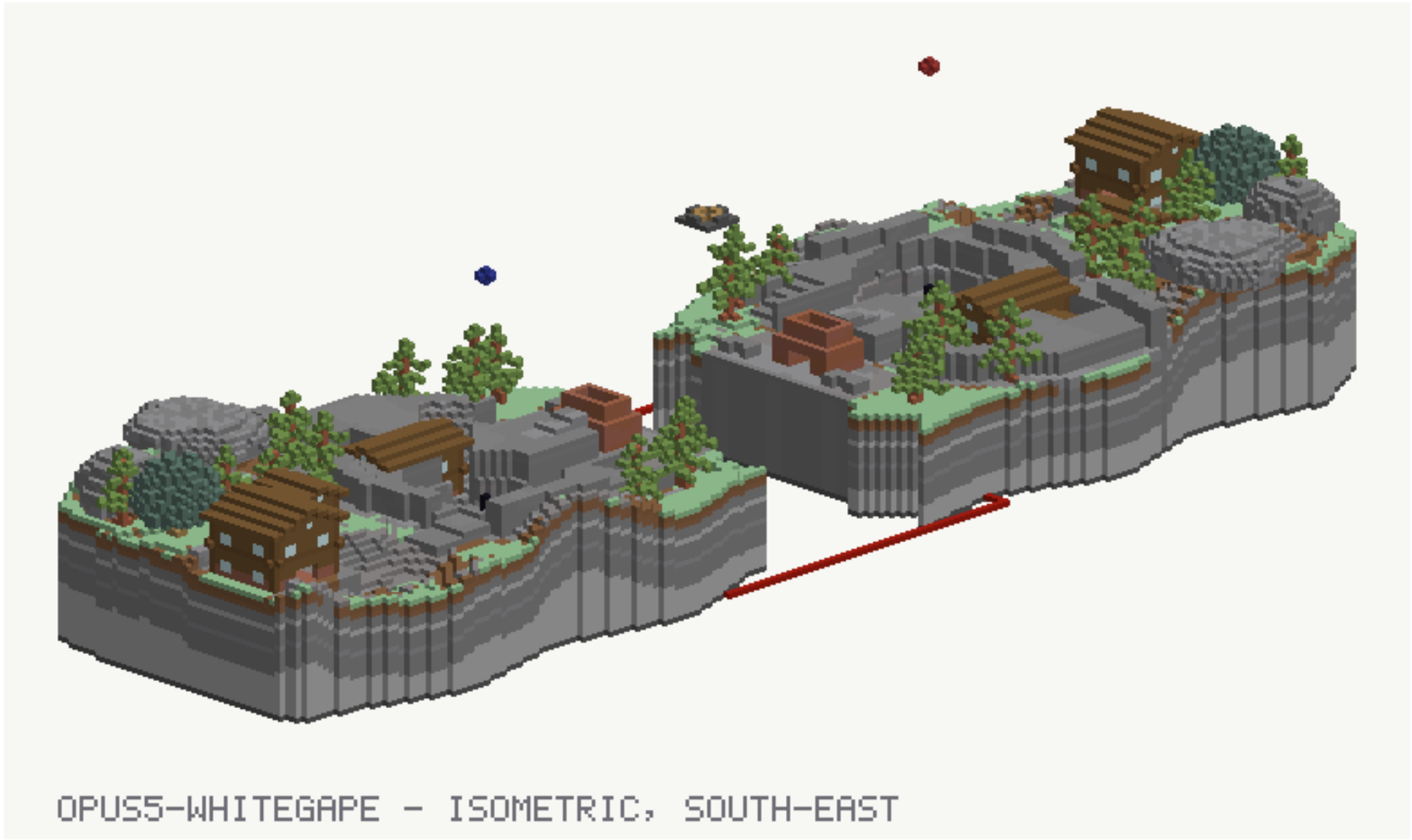
The rules are most of the studio

183 of them, over 28 families, each able to stop a request and say why. Not validation: the accumulated knowledge of what makes a map fail, written as things the software will not let happen.

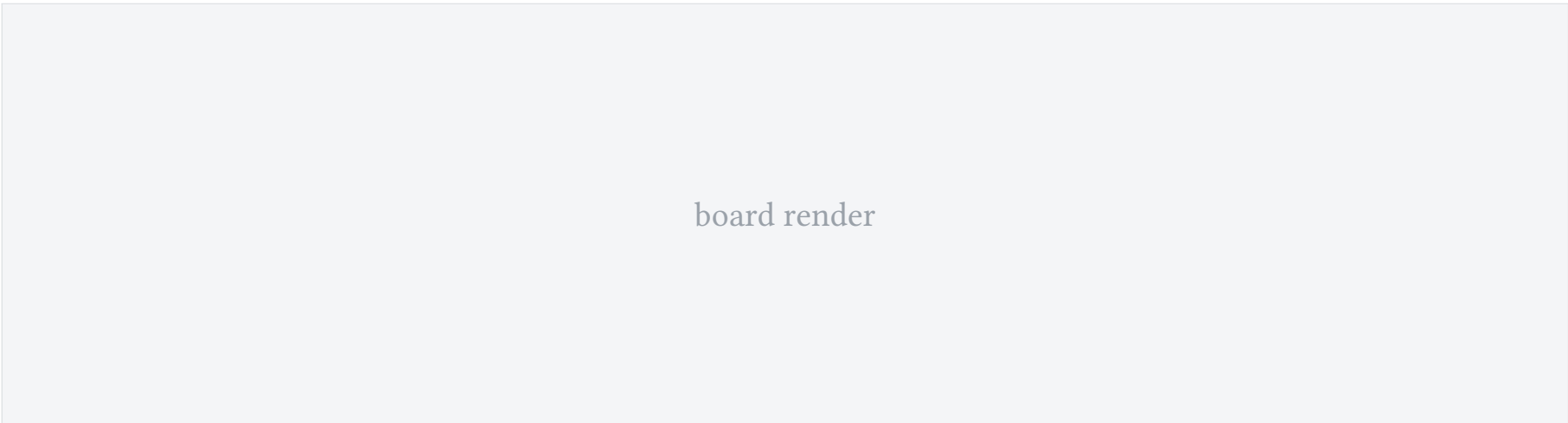
Family	About
SK 28	the sketch: shapes, layers, what drew no ground
DR 17	dressing: where a thing may rest, what it may not close off
PL 16	the plan: pieces, walls, spawns, objectives
WX 13	the room stamped for a spawn or a wool
OB 12	objectives: where a goal may stand, how a match can end
CT 10	the middle: the crossing, its holes, its build zones

A rule is not a constant. Many read *bands* whose values depend on what the board is played for, so three boards of experience on one objective teach nothing about the next.

Two boards made this way



Whitegape destroy, 14 a side
Authored from nothing, to no brief. A limestone gorge quarried from both rims, each team’s monument on the floor of its own pit.



Hushwater capture the wool, 18 a side
Pulled off the composer at seed 5 and then adapted. The arrangement is the studio’s; the place is the author’s.

What a session looks like

Every call each board’s author made was recorded. An authoring run is not a single request: it is hundreds of reads against a handful of writes, because an agent cannot see the map and every belief it holds is a number the studio handed back.

A 200 is not a yes. A request can succeed and still drop what it was sent, and the only place that is said is the warnings key. That is the most expensive lesson in the whole corpus.

Every figure on this sheet was drawn by the studio being described. The full paper, with both worked maps and the complete rule catalogue, is *How a PGM Map Gets Made*.